

Full-Stack Assignment

This assignment's purpose is to provide Algoscale with an example of an applicant's skills and capabilities as a full-stack developer.

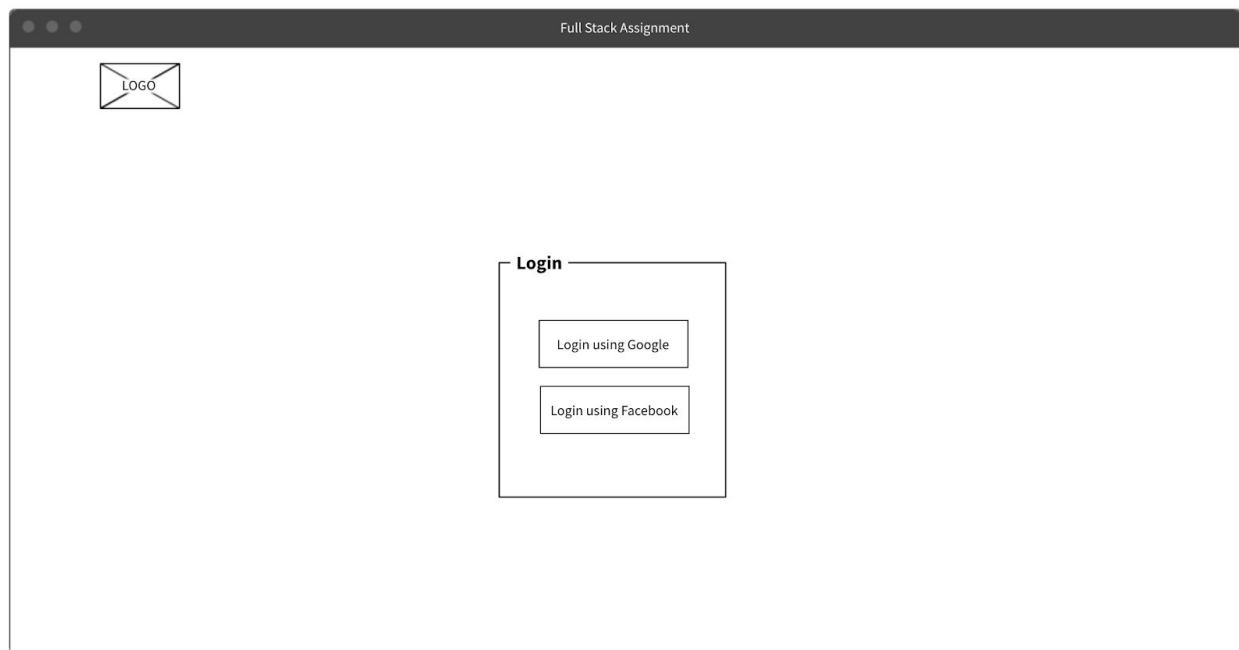
What needs to be done?

Build a full-stack single-page application (SPA) having 3 components as described below.

Login Page

Create a simple login page allowing users to pass through using their Google or Facebook accounts. Any package or existing module can be used to implement it.

After logging in, they should be taken to the "Contact Us" page (described in the next section) with a menubar having links to "Contact Us", "Analytics" page and logout



Contact Us Page

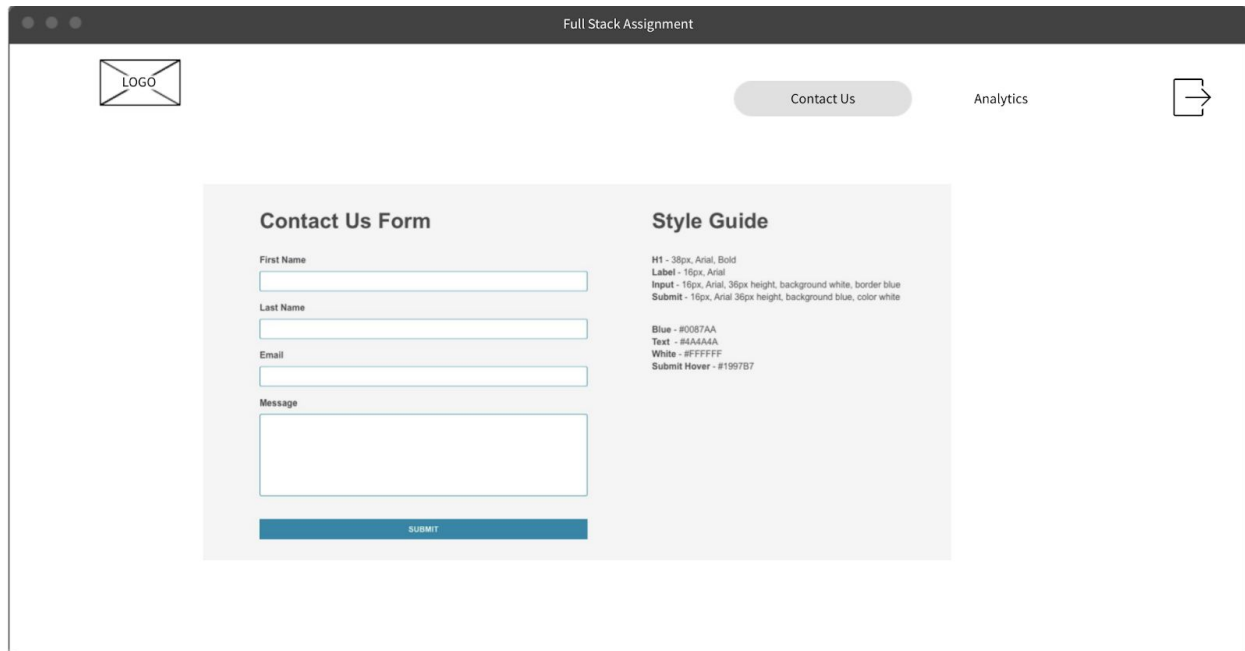
We would like potential applicants to create a simple "Contact us" form that saves a copy of the responses to a database.

This Contact us form will have the following 4 fields:

- First name

- Last name
- Email
- Message
- reCaptcha (below the Message box)

The first name, last name, and email fields will be text inputs, with the message being a text area. Be sure to add validation to the email field to make sure it's a valid email address, and prevent the form from submitting until a valid email address is entered.

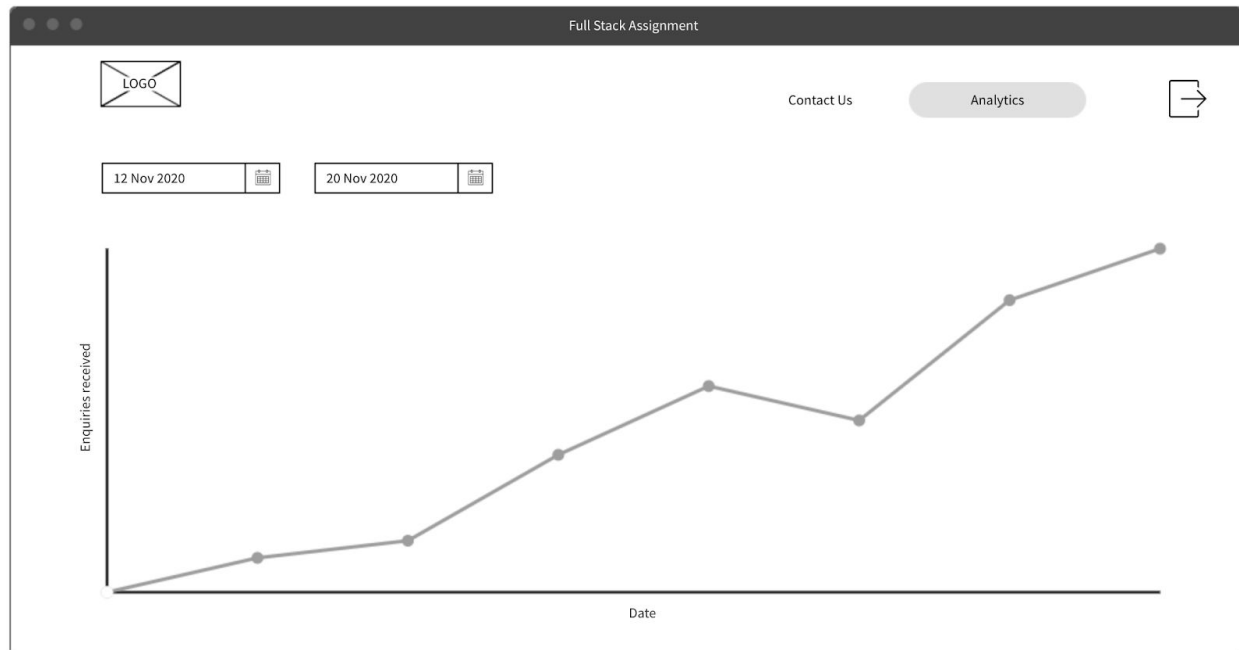


The mockup shows a web application window titled "Full Stack Assignment". In the top left corner is a placeholder for a logo labeled "LOGO". In the top right corner are two buttons: "Contact Us" and "Analytics", followed by a right-pointing arrow icon. The main content area is divided into two columns. The left column is titled "Contact Us Form" and contains four input fields: "First Name", "Last Name", "Email", and "Message" (a text area). Below these fields is a blue "SUBMIT" button. The right column is titled "Style Guide" and lists the following specifications:

- H1 - 36px, Arial, Bold
- Label - 16px, Arial
- Input - 16px, Arial, 36px height, background white, border blue
- Submit - 16px, Arial 36px height, background blue, color white
- Blue - #0077AA
- Text - #444444
- White - #FFFFFF
- Submit Hover - #1997B7

Analytics

- Date range
- Chart showing the number of form submissions from the contact us page form for every day between the selected dates



What can you use?

You are encouraged to use any framework of your choice on the backend as well as frontend among the modern frameworks available. In terms of the coding language, we prefer Python or Nodejs.

Also, you can pick any database of your choice. Some relevant ones are: MongoDB, ElasticSearch, MySQL. Though you must use a proper ORM layer to access the database.

What needs to be delivered?

- Frontend git repository
 - Login page (using single sign-on supporting Google/Facebook authentication)
 - Contact Us page as described above
 - Analytics page (after submitting contact page)
- Backend git repository
 - Set of APIs to enable the above functionality
- Screen recording of the working solution on your localhost with you describing the functionality in the form of a demo
- Architecture diagram of the solution

What would be evaluated?

- Complete functionality of the application
- Code quality

- Attention to detail on the UI and following the instructions
- Use of best practices (like lint)
- Dockerization (recommended)
- Test cases (bonus)

All the best!